

Conceptual Answers – Order Flow Imbalance Task

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1. What’s the motivation behind measuring OFI at multiple depth levels of the order book?

Measuring OFI at multiple levels of the limit order book enables a more complete understanding of the underlying liquidity dynamics of the asset. While best-level OFI captures changes at the top of the book, it overlooks pressure accumulating at deeper price levels, where participants may be placing larger but less aggressive orders. By continuing the OFI calculation beyond level 1, we are able to account for a broader range of activity that can still influence short-term price movement. Aggregating these observations across depths and applying principal component analysis allows us to derive a single, high-variance signal that summarizes the directional imbalance in the order book more effectively than any single level alone.

2. Why do the authors use Lasso regression rather than OLS for estimating cross-impact?

The authors choose Lasso regression instead of Ordinary Least Squares (OLS) to estimate cross-impact primarily due to issues related to dimensionality and multicollinearity. Given that cross-impact modeling involves a large number of predictors (approximately 100 assets), the direct use of OLS would produce unstable estimates, as the number of parameters could exceed the number of observations, especially over shorter estimation intervals. Furthermore, significant correlations among cross-asset order flow imbalances (OFIs) further increase the instability in such OLS estimates. Lasso regression effectively mitigates these issues by introducing a regularization term that selects only the most significant predictors while setting negligible coefficients to zero. Thus, this method improves both the stability and

interpretability of the model by highlighting meaningful relationships between assets and avoiding overfitting.

3. Why is OFI considered a better predictor of short-term returns than trade volume?

Trade volume provides information about the magnitude of transactions but not about their directional pressure or aggressiveness. In contrast, OFI incorporates both the side (bid vs. ask) and the nature (add, cancel, or trade) of order book updates, capturing real-time supply and demand imbalances. This directional sensitivity makes OFI more informative for predicting short-term price movements. Empirically, OFI correlates more strongly with return fluctuations than raw volume, particularly in high-frequency settings.